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COURSE CODE: SBT-DF204
LAB NAME: INVESTIGATING HARASSMENT EMAIL TRAFFIC WITH WIRESHARK
DATE: 05/10/2026
INSTRUCTOR: AMINU IDRIS

This individual case study asks you to investigate a simulated harassment-email complaint using supplied email material, a class roster, and a packet capture. Your report must show how you preserved the evidence, found the relevant web activity in Wireshark, and assessed whether the traffic supports attribution to a student in Chemistry 109. Do not assume that an IP address proves identity. Your conclusion must follow the evidence and explain its limits.

Case scenario

Lily Tuckrige, a Chemistry Department teacher, reports repeated harassing emails sent to her personal Yahoo Mail account. Full headers from an earlier message point to an IP address in a shared student residence where an open wireless router is in use. On Monday 21 July, a later message arrived through the web service willselfdestruct.com. The service briefly displayed the message and then reported that it had been destroyed. The university captured traffic at the residence network boundary and asks you to determine whether the available evidence supports attribution to a student in Chemistry 109.

Case materials

Use the scenario-provided first email, full headers, second email, class roster, and network traffic capture. **Scenario page:** digitalcorpora.org/corpora/scenarios/nitroba-university-harassment-scenario

Traffic capture: digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap

If the source page also provides a published checksum, compare it with the hash you calculate. If it does not, record your calculated SHA-256 value in the report so another examiner can identify the evidence file you analysed.

What you must submit

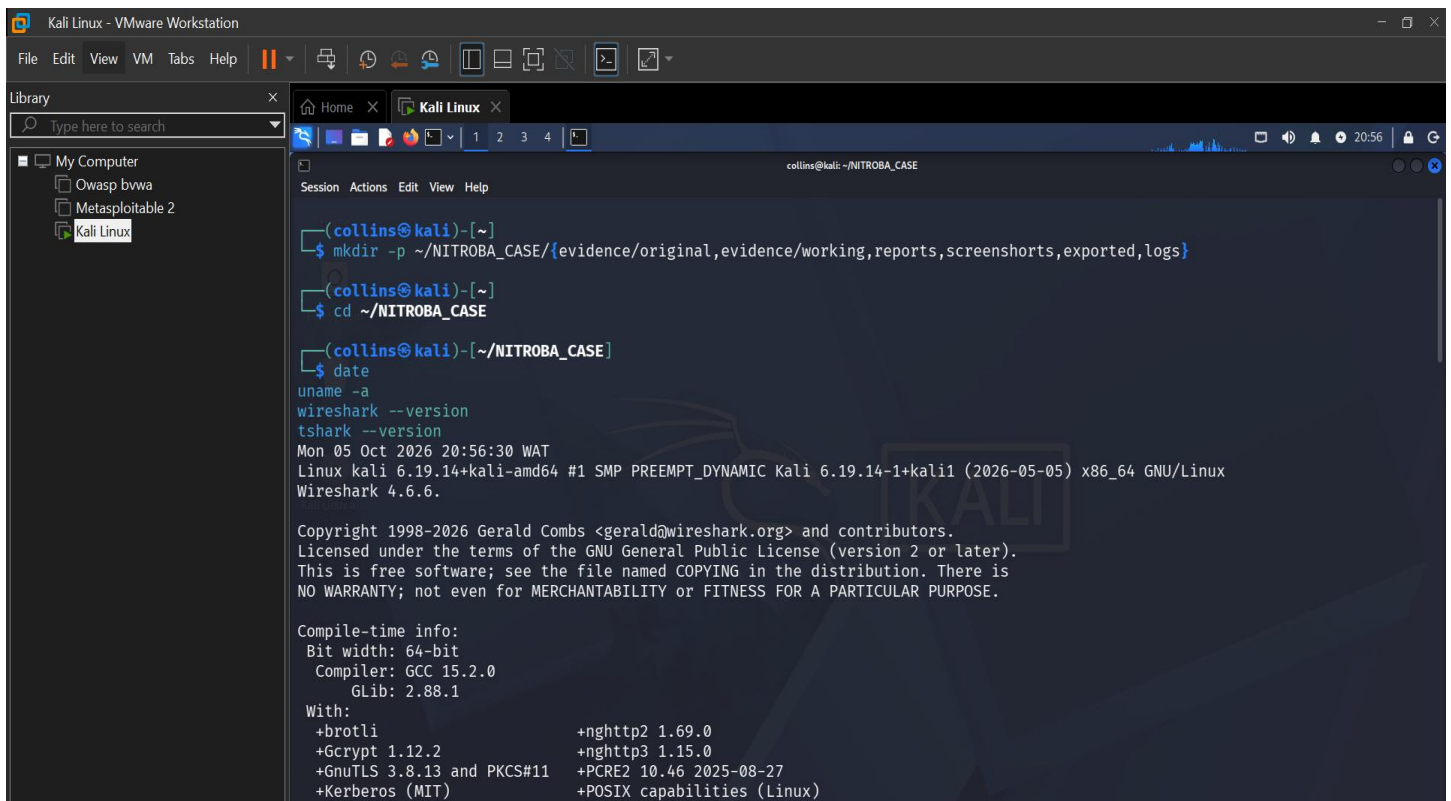
Item	Required output	What to provide
1	Forensic working copy	Download the scenario traffic capture and work only from your copied evidence file. Record the original file name, source link, file size, download date, and SHA-256 value.
2	Wireshark analysis	Use display filters, packet inspection, and Follow TCP Stream where appropriate. Preserve packet numbers and exact timestamps for every material finding.
3	Evidence-based conclusion	State only what the traffic supports. Separate direct observations, reasonable inferences, and limitations caused by the shared open wireless network.
4	Student report	Submit one report in PDF or DOCX format, with an evidence log and clearly labelled screenshots in an appendix or separate ZIP file.

Investigation task

Conduct a methodical forensic review of the packet capture. Use the questions below as the structure for your findings. For every material finding, provide the Wireshark display filter, packet number or stream reference, timestamp, and a short explanation of why the evidence matters. You may use the filters as starting points, but test and adapt them against the actual capture.

No	Investigation question	Minimum evidence to discuss
1	What file did you acquire, and how did you preserve its integrity?	Give the file name, source, size, SHA-256 value, and the method you used to create a working copy.
2	Which client system contacted the web service?	Identify the client IP and the service IP. Cite the packet number and filter that support each value.
3	What evidence links the client to the harassment message?	Show the relevant HTTP POST and explain how its content, form field, URI, or stream correlates with the reported message.
4	Which device made the relevant request?	Identify the MAC address associated with the client traffic and explain why a shared IP address alone cannot prove identity.
5	Who is associated with the device?	Use only scenario-provided cookie or browser evidence to link the device to an identity. State the packet evidence and avoid unsupported assumptions.
6	Is that person on the Chem 109 roster?	Compare the supported identity with the provided roster and show the basis for your answer.
7	When did the activity occur?	Create a short timeline using exact PCAP timestamps. State the time zone displayed by Wireshark and any conversion you make.
8	What conclusion can you defend?	Give a careful conclusion, confidence level, and at least two limitations or alternative explanations.

ANSWER:



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the following commands and output:

```
(collins@kali)-[~]
$ mkdir -p ~/NITROBA_CASE/{evidence/original,evidence/working,reports,screenshots,exported,logs}
(collins@kali)-[~]
$ cd ~/NITROBA_CASE
(collins@kali)-[~/NITROBA_CASE]
$ date
Mon 05 Oct 2026 20:56:30 WAT
$ uname -a
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux
$ wireshark --version
Wireshark 4.6.6.

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This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
  Bit width: 64-bit
  Compiler: GCC 15.2.0
  GLib: 2.88.1
  With:
    +brotli
    +Gcrypt 1.12.2
    +GnuTLS 3.8.13 and PKCS#11
    +Kerberos (MIT)
    +nghttp2 1.69.0
    +nghttp3 1.15.0
    +PCRE2 10.46 2025-08-27
    +POSIX capabilities (Linux)
```

```
Kerberos (MIT)                +POSIX capabilities (Linux)
+libnl 3                        +Qt 6.10.2
+libpcap                       +QtDBus
+libsmi 0.4.8                  +QtMultimedia
+libxml2 2.15.2                +Snappy 1.2.2
+Lua 5.4.8                     +xxhash 0.8.3
+LZ4 1.10.0                    +zlib 1.3.2
+MaxMind                       +Zstandard 1.5.7
+Minizip 1.3.2
Without:
-automatic updates -zlib-ng

Runtime info:
  OS: Linux 6.19.14+kali-amd64
  CPU: Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz (with SSE4.2)
  Memory: 4165 MB of physical memory
  GLib: 2.88.1
  Locale: LC_TYPE=en_NG
  Plugins: supported, 0 loaded
With:
+broccoli 1.2.0
+c-ares 1.34.6
+Gcrypt 1.12.2
+GnuTLS 3.8.13
+libpcap 1.10.6 (64-bit time_t, with TPACKET_V3)
+libsmi 0.4.8
+LZ4 1.10.0
+nghttp2 1.69.0
+nghttp3 1.15.0
+PCRE2 10.46 2025-08-27
+Qt 6.10.2
+xxhash 803
+zlib 1.3.2
+Zstandard 1.5.7
TShark (Wireshark) 4.6.6.

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Licensed under the terms of the GNU General Public License (version 2 or later).
This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
  Bit width: 64-bit
  Compiler: GCC 15.2.0
  GLib: 2.88.1
With:
+broccoli 1.2.0                +MaxMind
+Gcrypt 1.12.2                 +nghttp2 1.69.0
+GnuTLS 3.8.13 and PKCS#11     +nghttp3 1.15.0
+Kerberos (MIT)               +PCRE2 10.46 2025-08-27
+libnl 3                       +POSIX capabilities (Linux)
+libpcap                       +Snappy 1.2.2
+libsmi 0.4.8                  +xxhash 0.8.3
+libxml2 2.15.2                +zlib 1.3.2
+Lua 5.4.8                     +Zstandard 1.5.7
+LZ4 1.10.0
Without:
-zlib-ng

Runtime info:
  OS: Linux 6.19.14+kali-amd64
  CPU: Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz (with SSE4.2)
  Memory: 4165 MB of physical memory
  GLib: 2.88.1
  Locale: LC_TYPE=en_NG
  Plugins: supported, 0 loaded
With:
+broccoli 1.2.0
+c-ares 1.34.6
+Gcrypt 1.12.2
+GnuTLS 3.8.13
+libpcap 1.10.6 (64-bit time_t, with TPACKET_V3)
+libsmi 0.4.8
+LZ4 1.10.0
+nghttp2 1.69.0
+nghttp3 1.15.0
+PCRE2 10.46 2025-08-27
+xxhash 803
+zlib 1.3.2
+Zstandard 1.5.7

(collins@kali) - [~/NITROBA_CASE]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```

(collins@kali) ~ /NITROBA_CASE
$ cd ~/NITROBA_CASE/evidence/original

(collins@kali) ~ /NITROBA_CASE/evidence/original
$ wget -O nitroba.pcap \
'https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap'
--2026-10-05 21:02:53-- https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com) ... 52.92.168.129, 3.5.78.194, 16.15.40.248, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|52.92.168.129|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: 'nitroba.pcap'

nitroba.pcap          100%[=====>] 53.58M  469KB/s   in 1m 43s

2026-10-05 21:04:42 (534 KB/s) - 'nitroba.pcap' saved [56180821/56180821]

(collins@kali) ~ /NITROBA_CASE/evidence/original
$ ls -lh nitroba.pcap
file nitroba.pcap
-rw-rw-r-- 1 collins collins 54M Nov 22 2020 nitroba.pcap
nitroba.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 4096)

(collins@kali) ~ /NITROBA_CASE/evidence/original
$ sha256sum nitroba.pcap
2b77a9eafc1d6af163d1ba793c96dbccacb04e6befdfa0b01f8c67553ec2fb  nitroba.pcap

(collins@kali) ~ /NITROBA_CASE/evidence/original
$ md5sum nitroba.pcap
9981827f11968773ff815e39f5458ec8  nitroba.pcap

(collins@kali) ~ /NITROBA_CASE/evidence/original
$ md5sum nitroba.pcap
9981827f11968773ff815e39f5458ec8  nitroba.pcap

(collins@kali) ~ /NITROBA_CASE/evidence/original
$ sha1sum nitroba.pcap
65656392412add15f93f8585197a8998aaeb50a1  nitroba.pcap

(collins@kali) ~ /NITROBA_CASE/evidence/original
$ sha256sum nitroba.pcap > ~/NITROBA_CASE/logs/original_sha256.txt
md5sum nitroba.pcap > ~/NITROBA_CASE/logs/original_md5.txt
sha1sum nitroba.pcap > ~/NITROBA_CASE/logs/original_sha1.txt

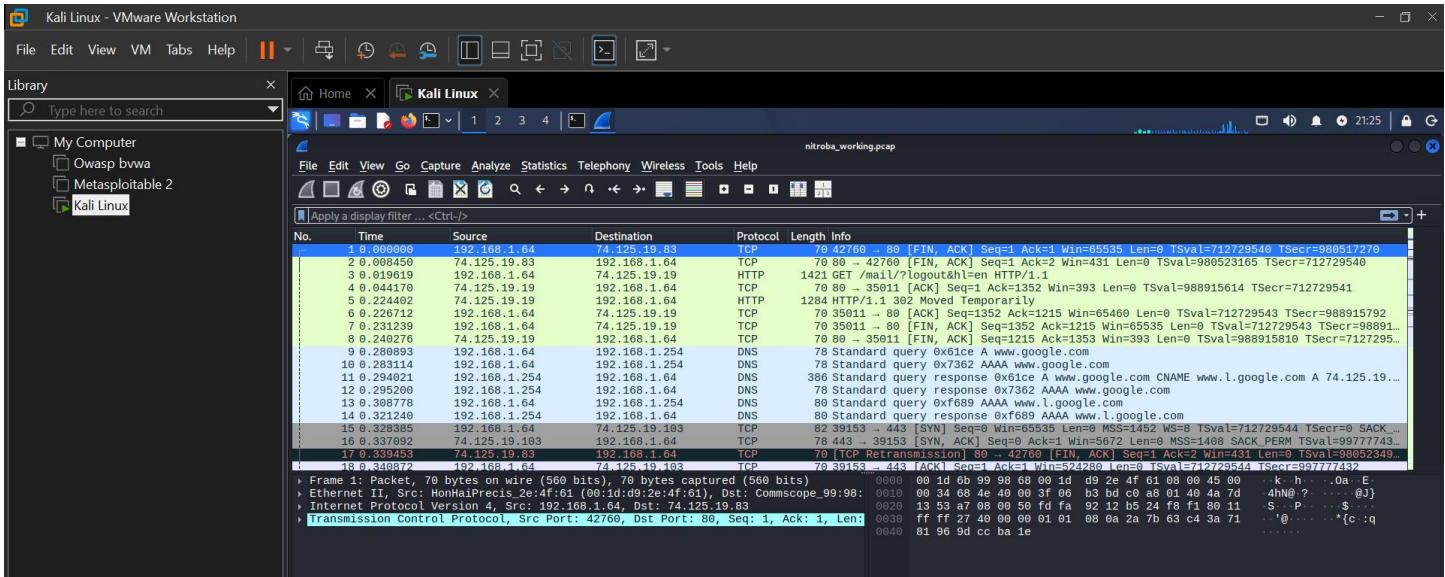
```

Wireshark starting points

The following filters support the workflow shown in the case material. Replace the values in angle brackets with values you derive from the capture. Do not treat a filter result as a conclusion until you inspect the packet details and document the connection to the investigation question.

Purpose	Example display filter
Locate web requests	http.request.method == "GET" && http.host == "www.willselfdestruct.com"
Search page content	data-text-lines contains "secure anonymous E-mail"
Identify form activity	http.request.method == "POST"
Narrow to the relevant exchange	ip.src == <client_ip> && ip.dst == <service_ip> && http.request.method == "POST"
Trace a device	eth.addr == <client_mac>
Look for identity evidence	eth.src == <client_mac> && http.cookie contains "@"

ANSWER:



Required report structure

1. Case reference and student identification. Include your full name, registration number, course code, and the case title.
2. Executive conclusion. State the supported conclusion in a short, cautious paragraph. Do not name a suspect without explaining the evidence link.
3. Evidence acquisition and integrity. Record the source, file name, size, hash, working copy, and any published checksum comparison.
4. Method. Explain the sequence of Wireshark filters, packet inspection, and stream review you used.
5. Findings. Answer each of the eight investigation questions with citations to packet numbers, timestamps, screenshots, and evidence-log IDs.
6. Timeline. Show the relevant sequence of requests and responses using the timestamps from the PCAP.
7. Attribution assessment. Distinguish observed facts from inferences, give a confidence level, and explain the shared-network limitation and any other alternative explanation.
8. Appendices. Include labelled screenshots and the completed evidence log. Screenshots must show the active filter, packet list, relevant packet detail, and time where possible.

ANSWER:

```
(collins@kali)-[~/NITROBA_CASE/evidence]
└─$ sha256sum original/nitroba.pcap
sha256sum working/nitroba_working.pcap
2b77a9eaeefc1d6af163d1ba793c96dbccacb04e6befdfa0b01f8c67553ec2fb original/nitroba.pcap
2b77a9eaeefc1d6af163d1ba793c96dbccacb04e6befdfa0b01f8c67553ec2fb working/nitroba_working.pcap

(collins@kali)-[~/NITROBA_CASE/evidence]
└─$ cmp original/nitroba.pcap working/nitroba_working.pcap

(collins@kali)-[~/NITROBA_CASE/evidence]
└─$ cd ~/NITROBA_CASE/evidence/working

capinfos nitroba_working.pcap
File name: nitroba_working.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 4096 bytes
Number of packets: 94 k
File size: 56 MB
Data size: 54 MB
Capture duration: 15759.950751 seconds
Earliest packet time: 2008-07-22 02:51:07.095278
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256: 2b77a9eaeefc1d6af163d1ba793c96dbccacb04e6befdfa0b01f8c67553ec2fb
SHA1: 65656392412add15f93f8585197a8998aaeb50a1
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 4096
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 94410

(collins@kali)-[~/NITROBA_CASE/evidence/working]
└─$ capinfos nitroba_working.pcap \
> ~/NITROBA_CASE/logs/capinfos.txt

(collins@kali)-[~/NITROBA_CASE/evidence/working]
└─$ wireshark ~/NITROBA_CASE/evidence/working/nitroba_working.pcap
** (Wireshark:17301) 21:24:07.734424 [GUI WARNING] -- Failed to register with host portal QDBusError("org.freedesktop.portal.Error.Failed", "Could not register app ID: Connection already associated with an application ID")
^C

(collins@kali)-[~/NITROBA_CASE/evidence/working]
└─$ TZ=UTC tshark -r ~/NITROBA_CASE/evidence/working/nitroba_working.pcap \
-T fields \
-e frame.number \
-e frame.time \
-c 10
1      2008-07-22T01:51:07.095278000+0000
2      2008-07-22T01:51:07.103728000+0000
3      2008-07-22T01:51:07.114897000+0000
4      2008-07-22T01:51:07.139448000+0000
5      2008-07-22T01:51:07.319680000+0000
6      2008-07-22T01:51:07.321990000+0000
7      2008-07-22T01:51:07.326517000+0000
8      2008-07-22T01:51:07.335554000+0000
9      2008-07-22T01:51:07.376171000+0000
10     2008-07-22T01:51:07.378392000+0000

(collins@kali)-[~/NITROBA_CASE/evidence/working]
└─$
```

Evidence log template

Include a completed version of this table in your report or evidence appendix. You may add rows where needed.

Evidence ID	Packet or item	Finding	Why it matters	Screenshot or appendix
E01	nitroba.pcap	SHA-256 matches published value	Establishes evidence inte	
E02	IP packets	192.168.15.4 communicates with relevant network/service	Identifies client	
E03	Frame 83601	HTTP POST to willselfdestruct.com	Directly links client traffic to reported web service	
E04	Frame 83601/ethernet II	MAC 00:17:f2:e2:c0:ce	Identifies observed network device	
E05	Gmail HTTP/cookies packet	jcoachj@gmail.com	Identity/session correlation	
E06	Chem109 roster	Johnny coach	Correlates identifier to roster member	